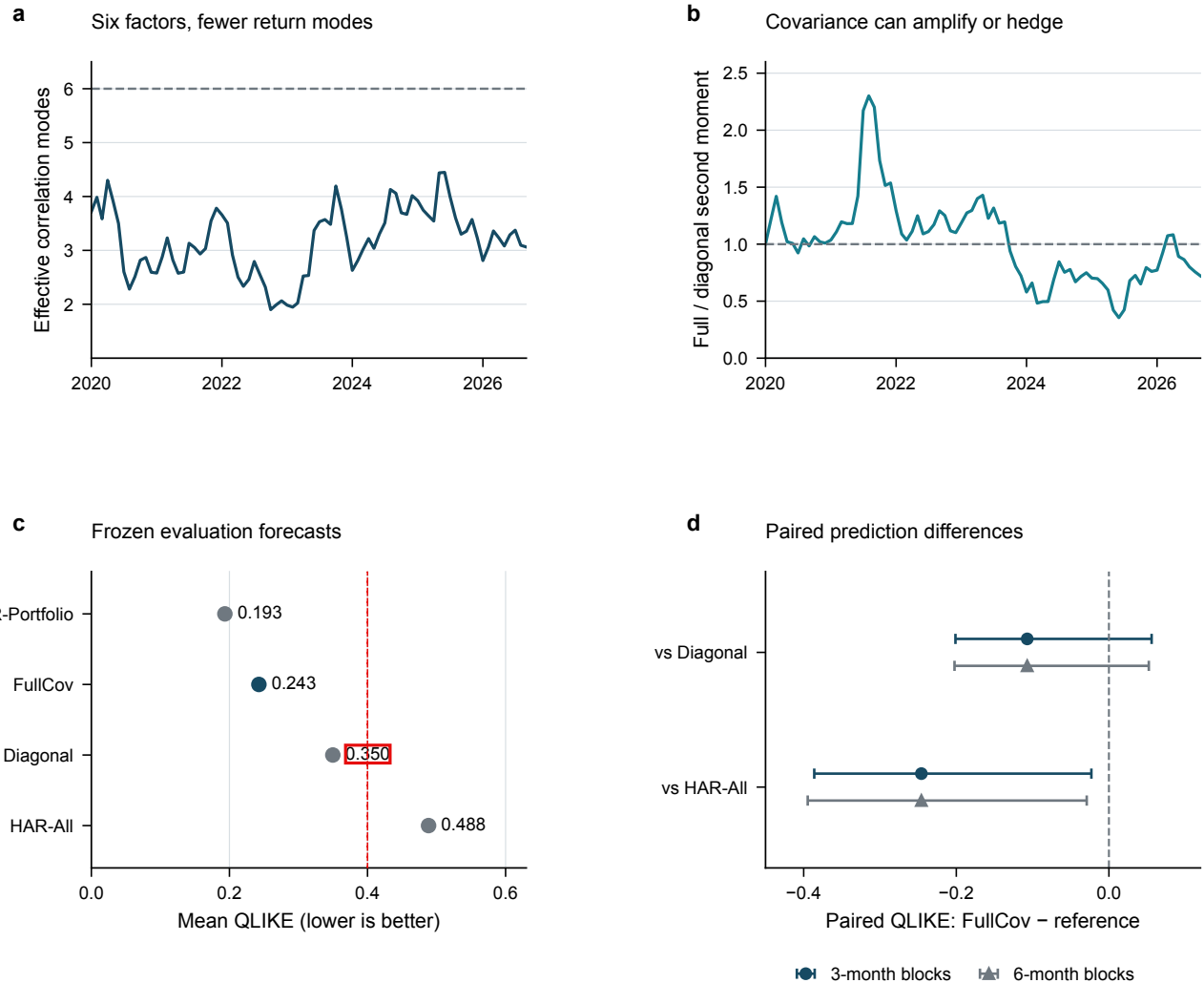
a
Six factors, fewer return modes
Effective correlation modes
6
5
4
3
2
2020
2022
2024
2026
b
Covariance can amplify or hedge
Full / diagonal second moment
2.5
2.0
1.5
1.0
0.5
0.0
2020
2022
2024
2026
c
Frozen evaluation forecasts
HAR-Portfolio
0.193
FullCov
0.243
Diagonal
0.350
HAR-All
0.488
0.0
0.2
0.4
0.6
Mean QLIKE (lower is better)
d
Paired prediction differences
vs Diagonal
vs HAR-All
−0.4
−0.2
0.0
Paired QLIKE: FullCov − reference
3-month blocks
6-month blocks